Samuel A. Lindwall

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Education

Georgia Institute of Technology | Atlanta, GA *August 2026 – Present*
Bachelor of Science in Computer Engineering Expected May 2028

Emory University | Atlanta, GA *August 2023 – May 2026*
Bachelor of Arts in East Asian Studies GPA: 3.95

Skills

Programming Languages: C, Python, Java, Verilog, RISC-V, MATLAB
Embedded Systems: STM32, ARM Cortex-M, PWM, UART, I^2C, SPI, DMA, PID
Tools: STM32CubeIDE, STM32CubeMX, logic analyzer, oscilloscope, soldering, Git, Linux
AI & Computer Vision: YOLO, DSPy, Agno, MLflow, RAG

Projects

Vision-Based Automatic Tracking Robot | C, STM32, UART, I^2C, PID *Summer 2026*

- Developed bare-metal C firmware for an STM32 Nucleo F446RE to control a pan-tilt hand tracking robot, running a 160 Hz PID loop using absolute encoder feedback over I^2C
- Implemented vision-based tracking on a Raspberry Pi using Pi Camera 2 and YOLO hand detection, sending target pan/tilt angles to the STM32 over UART for closed-loop motor control
- Debugged a timing mismatch between hand detection and encoder sampling by estimating the camera's position at the exact moment each frame is captured, eliminating hunting without adding a delay
- Reduced steady-state error from 4° to 1° by adding measured friction feedforward
- Increased motor engagement during tracking from 38% to 98% by gating the firmware deadband on a target velocity, eliminating a stop-start lurch cycle

"The Lick" Melody Generator | Verilog, FPGA, FSM *Fall 2025*

- Designed a finite-state machine in Verilog to sequence musical notes and control a piezo buzzer on a Basys2 FPGA
- Implemented clock dividers and counters using a 50 MHz system clock to generate square-wave frequencies and control note durations with 0.2 s timing intervals

Experience

Kashmir World Foundation | Remote *May – August 2026*
Software Engineer Intern

- Designed a Python sensor agent with Agno to control a gimbaled camera through natural-language commands on a fixed-wing UAV in ArduPilot SITL, using MAVLink ROI control
- Reduced command execution latency by approximately 90% with a deterministic tool-calling pipeline instead of Agno's built-in agent loop; validated on 50 test cases and benchmarked 15 models for cost, latency, and accuracy

Collins Aerospace (RTX) | Cedar Rapids, IA *May – August 2025*
Software Engineer Intern

- Developed a Python-based avionics test application integrating Llama 3.3, RAG, and deterministic tool-calling to generate flight-plan modifications and simulate abnormal flight scenarios
- Evaluated model reliability across 100 simulation scenarios using MLflow and systematic testing, achieving 85% output accuracy

Relevant Coursework

Programming HW/SW Systems; Signal Processing; Signals and Systems; Feedback Control Systems; Digital Electronics/FPGAs; Circuit Analysis; Data Structures & Algorithms; Object Oriented Programming

Activities

Autobots Research Group | Undergraduate Researcher *August 2026 – Present*

- Research and design algorithms for robots navigating dynamic environments

Marine Robotics Club | Firmware Engineer *August 2026 – Present*

- Develop embedded firmware for an autonomous surface-water robot, integrating sensor interfaces, actuator control, and communication systems for competitive robotic navigation